Markus Gruber, M.Sc.

Senior Cloud Infrastructure
DevOps Engineer

CI/CD Automation · *Release Optimization* · *Platform Engineering Specialist*

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Professional Profile



Senior Cloud Infrastructure & DevOps Engineer with 9+ years of experience automating build pipelines, optimizing container workloads, and maintaining high-availability software platforms. Holds an M.Sc. in Scientific Computing. Resilient problem solver resolving critical production bottlenecks. Specializes in reducing release cycle times, establishing robust telemetry/monitoring, and unifying development and ops workflows through infrastructure as code and automated release verification.

Engineering Approach & Mindset

- **Systematic Problem Solving:** First-principles analytical approach to systematic debugging and stability optimization.
- **Quality Efficiency Focus:** Pragmatic engineering mindset balancing rapid deployment with long-term maintainability and high performance.
- **Cross-Functional Integration:** Acts as the technical glue bridging development, QA, and architecture.

Engineering Profile

TECHNICAL FOUNDATION & PLATFORM / DEVOPS DOMAINS

CI/CD & Automation	Container & Cloud	Observability & Systems
DevOps & GitLab CI/CD	Docker & Containerization	Telemetry & Profiling Suites
Automated Testing Pipelines	AWS & Azure Cloud Services	Linux Administration & Bash
Release Performance Testing	Infrastructure Automation	Python & C++ Integration

CORE TECHNICAL SPECIALIZATION – PLATFORM ENGINEERING

- **CI/CD Pipeline Automation:** Designing scalable build and deployment pipelines, reducing release testing overhead and accelerating time-to-market.
- **Containerization & Cloud Operations:** Managing Docker container orchestration, environment standardization, and cross-platform infrastructure stability.
- **System Profiling & Performance Optimization:** Root-cause analysis of build bottlenecks, telemetry tracking, and automating release verification suites.

Professional Experience

Mindbreeze GmbH – Software / Quality Engineer (C++/Python) 09/2022 – 07/2026
C++, Python, REST APIs, Linux, Docker, DevOps, Telemetry & Performance Profiling | Team: 6 Engineers

- Engineered Python/Bash automation frameworks, reducing release performance test duration from 10 working days down to 3 working days.
- Optimized performance-critical backend components for enterprise architectures handling distributed data processing.
- Refactored search engine core modules and optimized index data pipelines to improve high-throughput query execution.
- Primary technical troubleshooter during release phases, resolving memory leaks and performance degradations in live systems.

Untis GmbH – Software Engineer (C++) 02/2022 – 06/2022
C++, System Refactoring, Performance Profiling, Core Desktop Modules | Team: 10 Engineers

- Refactored legacy C++/MFC core modules using Visual Studio to eliminate technical debt, optimize memory allocation, and resolve production crashes.

logi.cals – Embedded Software Engineer (C/C++) 02/2021 – 01/2022
C, C++, Embedded Systems, Safety-Critical Architectures, Deterministic Execution | Team: 6 Engineers

- Engineered deterministic C/C++ runtime execution systems (Software PLCs) and automated verification test suites for safety-standard certification across hardware platforms.

EYYES – Software Engineer (C++/Python) 04/2017 – 12/2020
C++, Python, Real-Time Execution Pipelines, CUDA, Embedded GPU Edge | Team: 8 Engineers

- Developed computer vision and deep learning pipelines (Keras, ONNX) for industrial and embedded camera systems to ensure continuous real-time edge performance.
- Engineered low-latency C++ and CUDA execution routines for real-time video stream processing on embedded GPU edge devices.
- Implemented hardware-near software components in C++ to control camera sensors, stepper motors, and SerDes interfaces.
- Optimized detection accuracy and recognition precision by training multiple TensorFlow classifiers to 99.99% accuracy through rigorous hardware-near algorithm tuning and automated benchmarking suites.

Selected Software Projects

- **AWS Cloud Deep Learning Pipeline:** Cloud-native MLOps architecture featuring automated CI/CD ingestion pipelines and containerized PyTorch inference (GitHub).
  Tech: AWS, Docker, Python, PyTorch, MLOps, CI/CD
- **Azure AI Document Analyzer & RAG Infra:** Scalable cloud retrieval infrastructure on Azure utilizing automated deployment and vector database services (GitHub).
  Tech: Azure, Docker, Vector DBs, REST APIs, Infrastructure
- **Magnum C++ Game of Life:** Cross-platform build automation and high-performance benchmarking suite for C++ grid simulation (GitHub).
  Tech: Modern C++, CMake, Build Systems, Performance Profiling

Technical Skills

DevOps & Tooling: GitLab CI/CD, DevOps, GitHub Actions, Docker, Kubernetes, Git, Telemetry, Linux Administration.
Scripting & Systems: Python, Bash, Modern C++, C, SQL, Parallel Algorithms, Multithreading.
Cloud Platform: Amazon AWS (S3, RDS, EC2), Microsoft Azure, Linux (Ubuntu/Debian), Microservices, REST APIs.
Data & Infrastructure: PostgreSQL, DuckDB, Apache Superset, CUDA, OpenCL.

Education

M.Sc. Scientific Computing – University of Vienna 10/2013 – 06/2017
Thesis: Silent Data Corruption in FT-GMRES (Link, Advisor: Prof. Dr. W. Gansterer).

B.Sc. Industrial Simulation – FH St. Pölten 09/2010 – 06/2013
Thesis: Simulation of Nanocomposite Materials (Link, Advisor: Prof. Dr. T. Schrefl).

Languages & Work Status

German: Native (C2) · **English:** Full Professional Proficiency (C1/C2) · **Work Permit:** EU Citizen / Austria